Strategic Patient Risk Stratification

Optimizing Outcomes to Mitigate the HRRP Penalty Threat

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Introduction & Data Foundation

The Financial Threat

~ $1.5M

Annual Risk Exposure

3% penalty on all Medicare reimbursements

18% Raw Rate

Initial assessment

11.47% Verified

HRRP penalty group

The Business Mandate

Target : Strictly <30 Day Readmissions

Risk. : 3% Penalty on Medicare Reimbursements

Impact. : Approximately $1.5 Million Annual Financial Exposure

Hospital Readmissions Reduction Program (HRRP) mandates operational excellence in post-discharge care coordination

Rigorous Data Hygiene

Raw Encounters

101,766 initial records

Deceased Removed

1,616 patients excluded (cannot readmit)

Columns Dropped

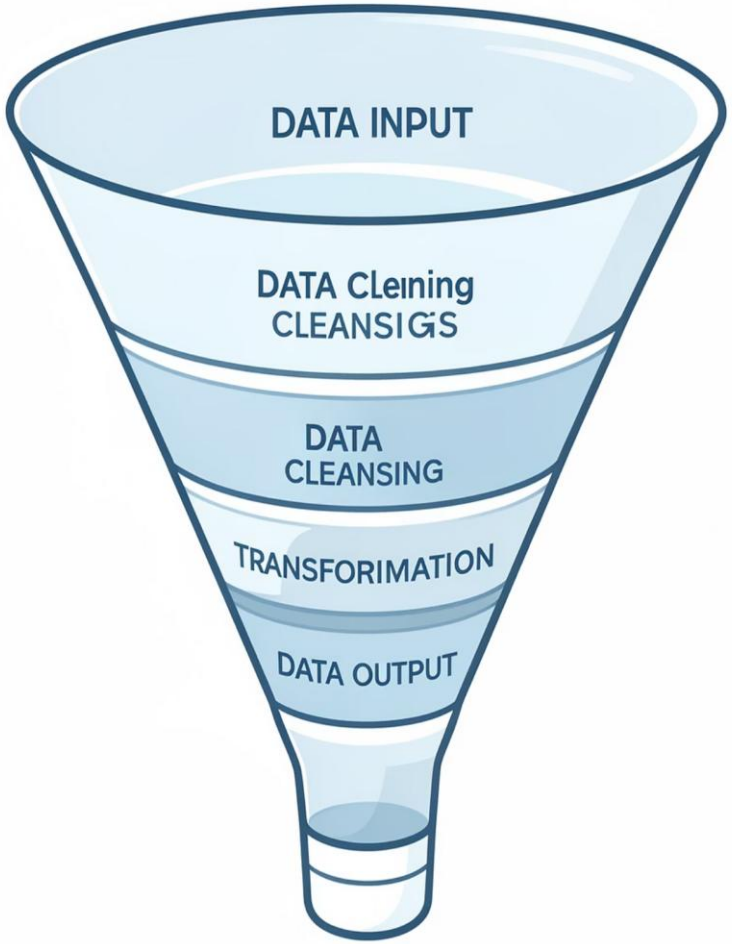
Weight (97% missing data), Payer Code (40% missing data)

Rows/Duplicates Checked

Minor missing data rows dropped 0 duplicates found

Clean Cohort

96,437 analytical records



Systematic data sanitation ensures analytical integrity and predictive validity

Demographic & Landscape

The "Needle in the Haystack"

Target Class

Only 11.47% of population represents <30 day readmissions readmissions

The Challenge

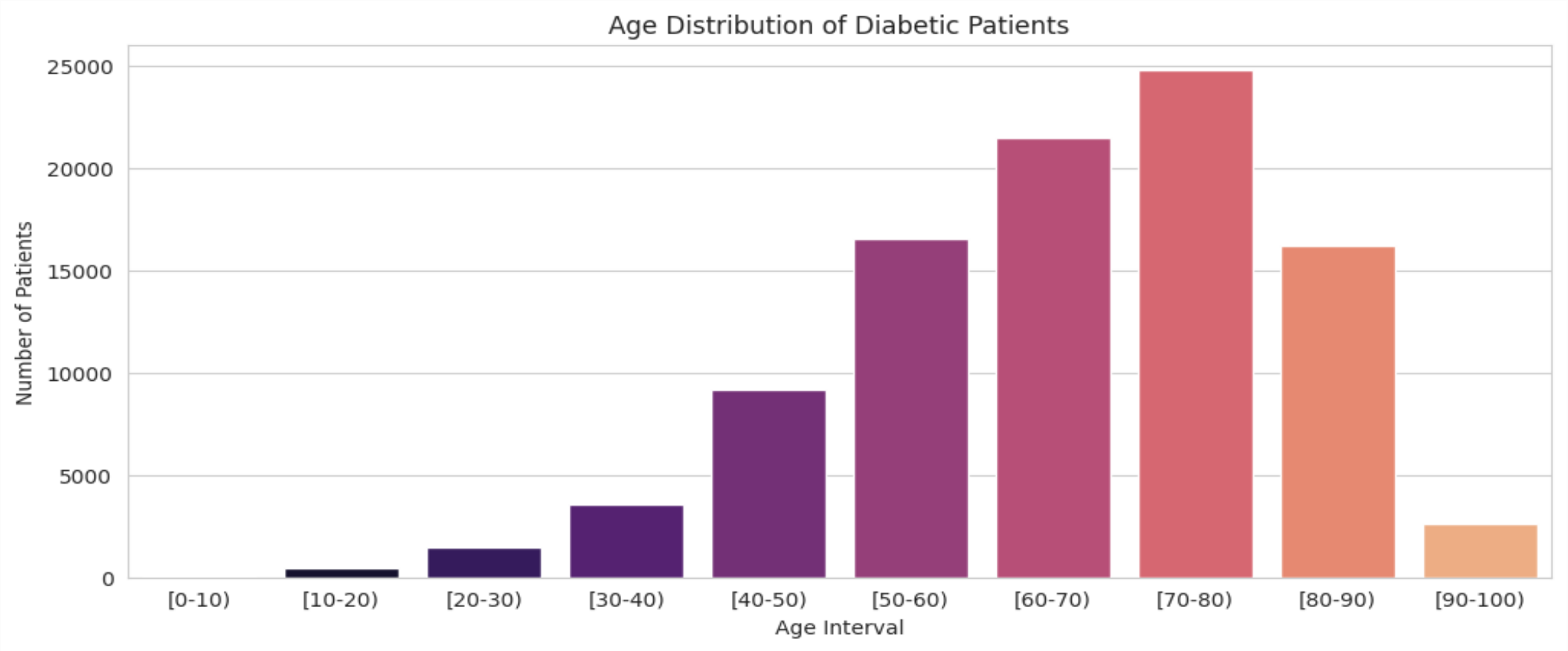
Predicting "No Readmission" achieves 89% accuracy but provides zero clinical value

Mission Critical

We must hunt for the minority minority signal within massive massive data noise

A Geriatric Challenge

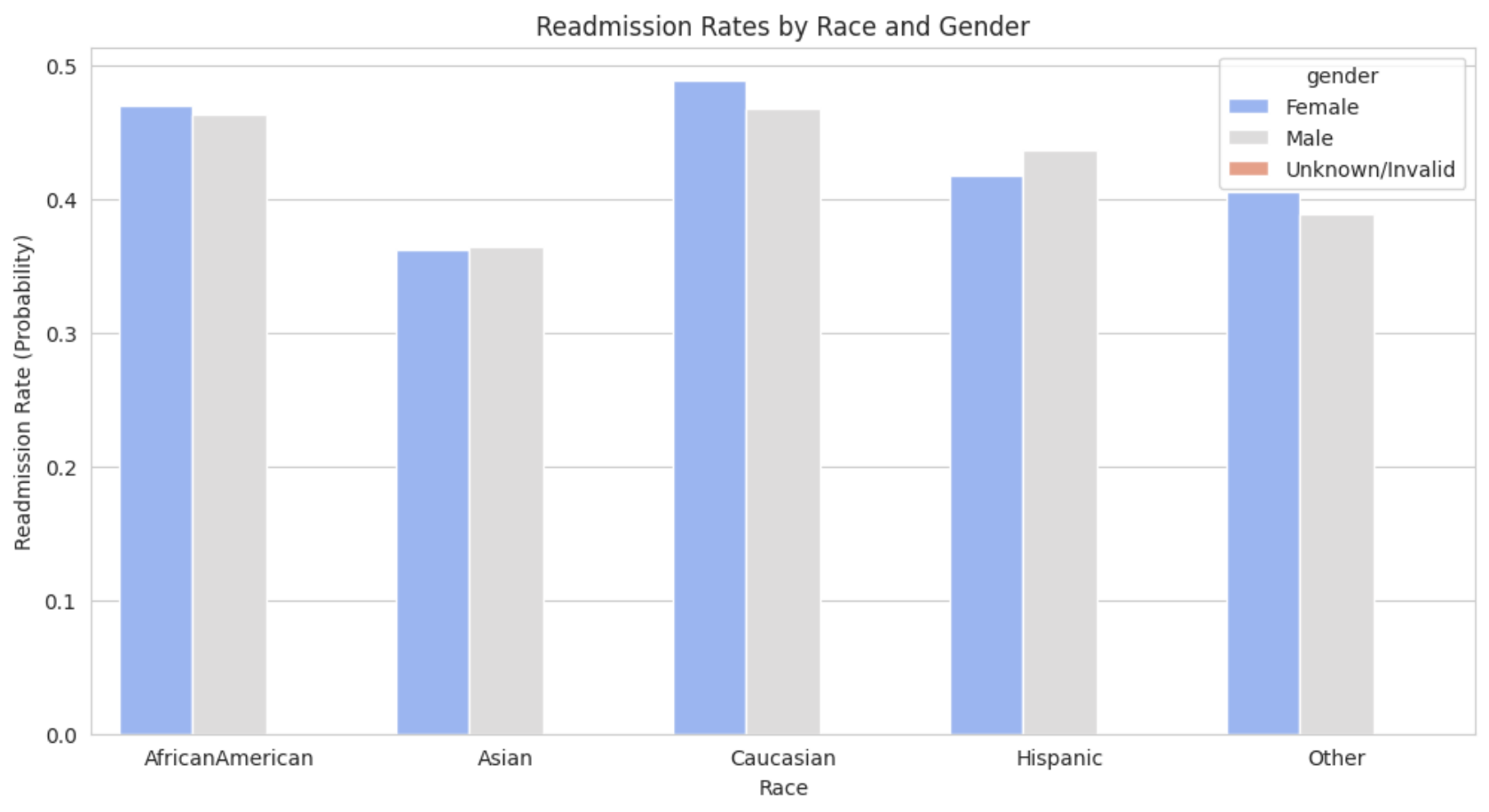
Understanding our patient demographic for targeted intervention design



Key Insight - Patient volume peaks dramatically between 70-80 years of age, representing our highest-risk cohort.

Demographic Risk Factors

Minor statistical variance in readmission rates by race suggests deeper systemic drivers



Root Cause Analysis

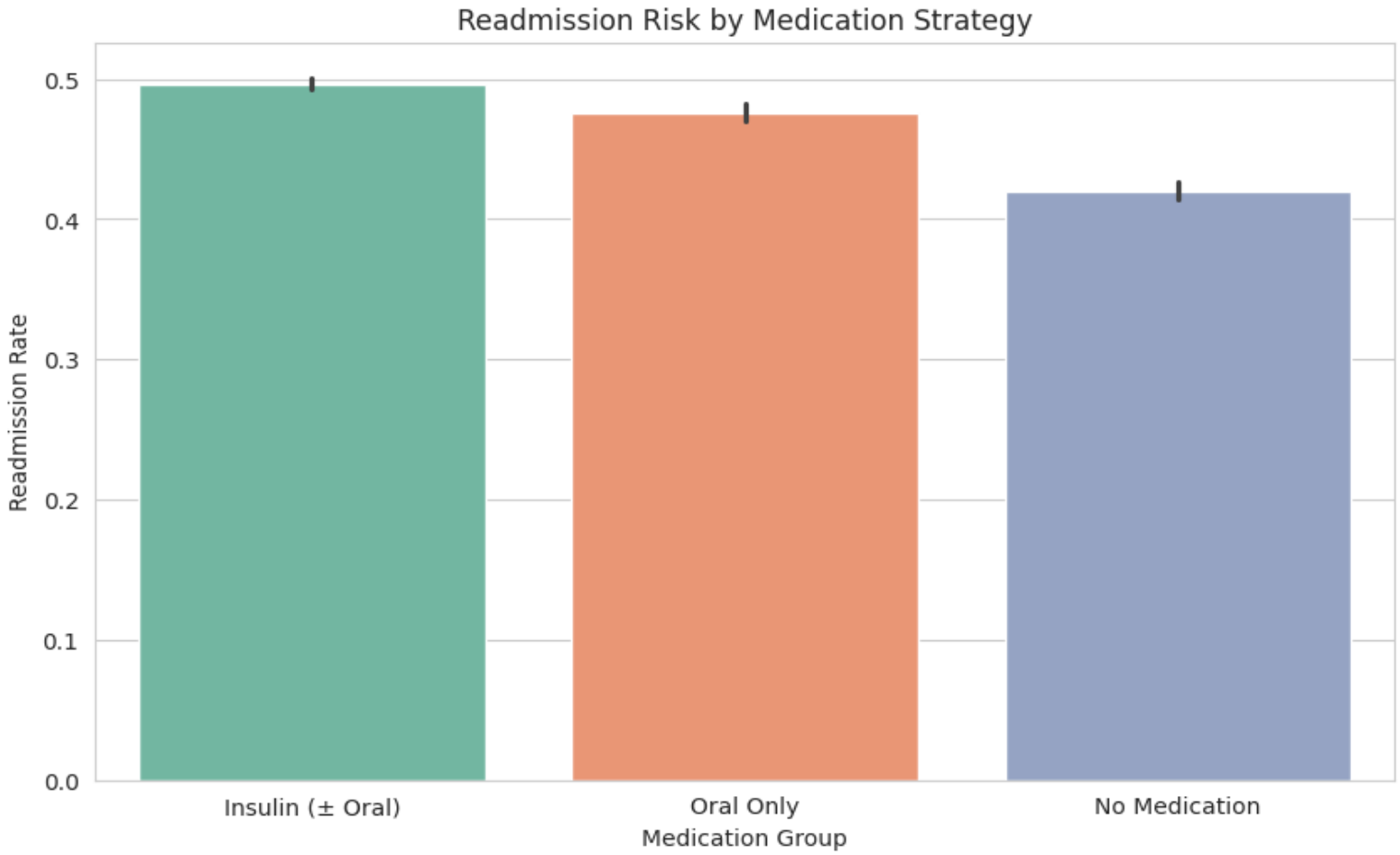
Observed differences are likely driven by **Social Determinants of Health** rather than biological factors:

- Transportation access barriers limiting follow-up up appointments
- Medication cost constraints constraints affecting adherence
- Language barriers in discharge instruction comprehension
- Food insecurity impacting impacting dietary compliance

Clinical & Operational EDA

The Insulin Risk Signal

Insulin users demonstrate dramatically elevated readmission probability



Insulin users have a **~49%** Total Readmission Probability.

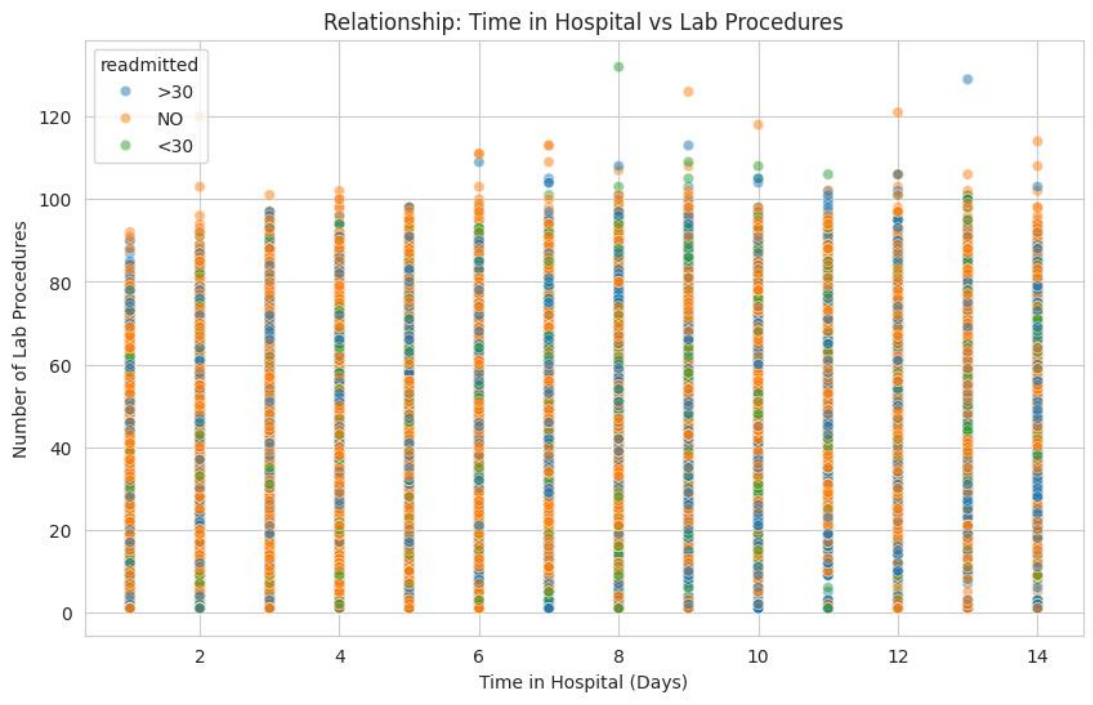
Critical Gap

Current 15-minute education protocol is insufficient for insulin management complexity

Intensity Does Not Equal Safety

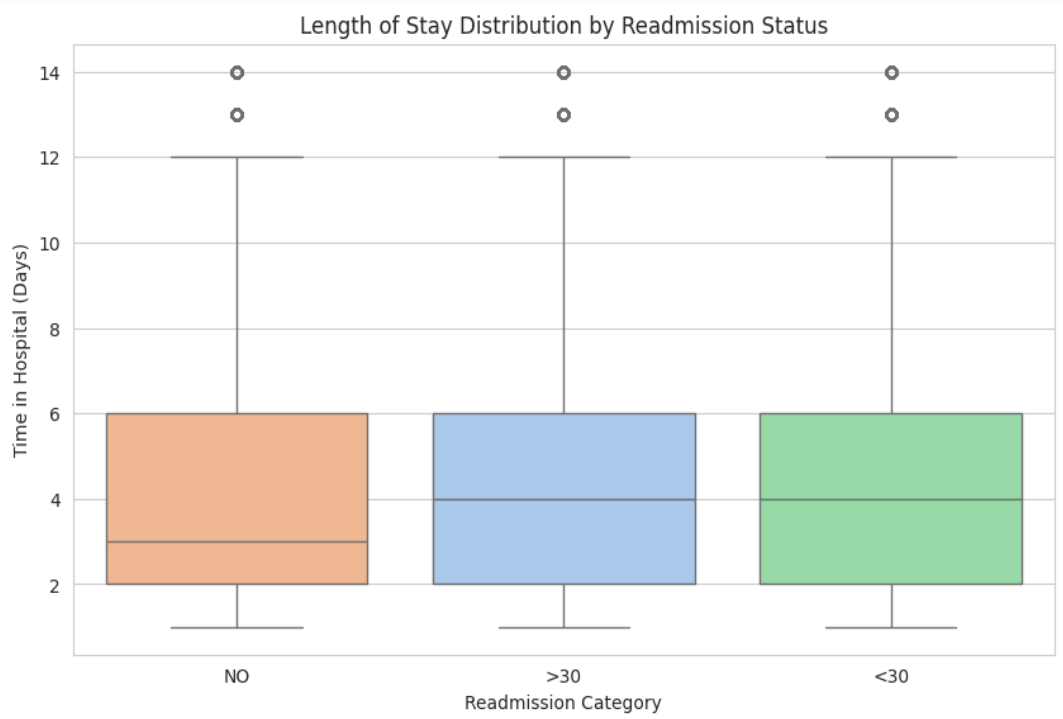
Analyzing the trade-off between recourses utilization and patient outcomes

RESOURCE UTILIZATION



Positive Correlation: Longer hospital stays significantly increase lab procedure volume.

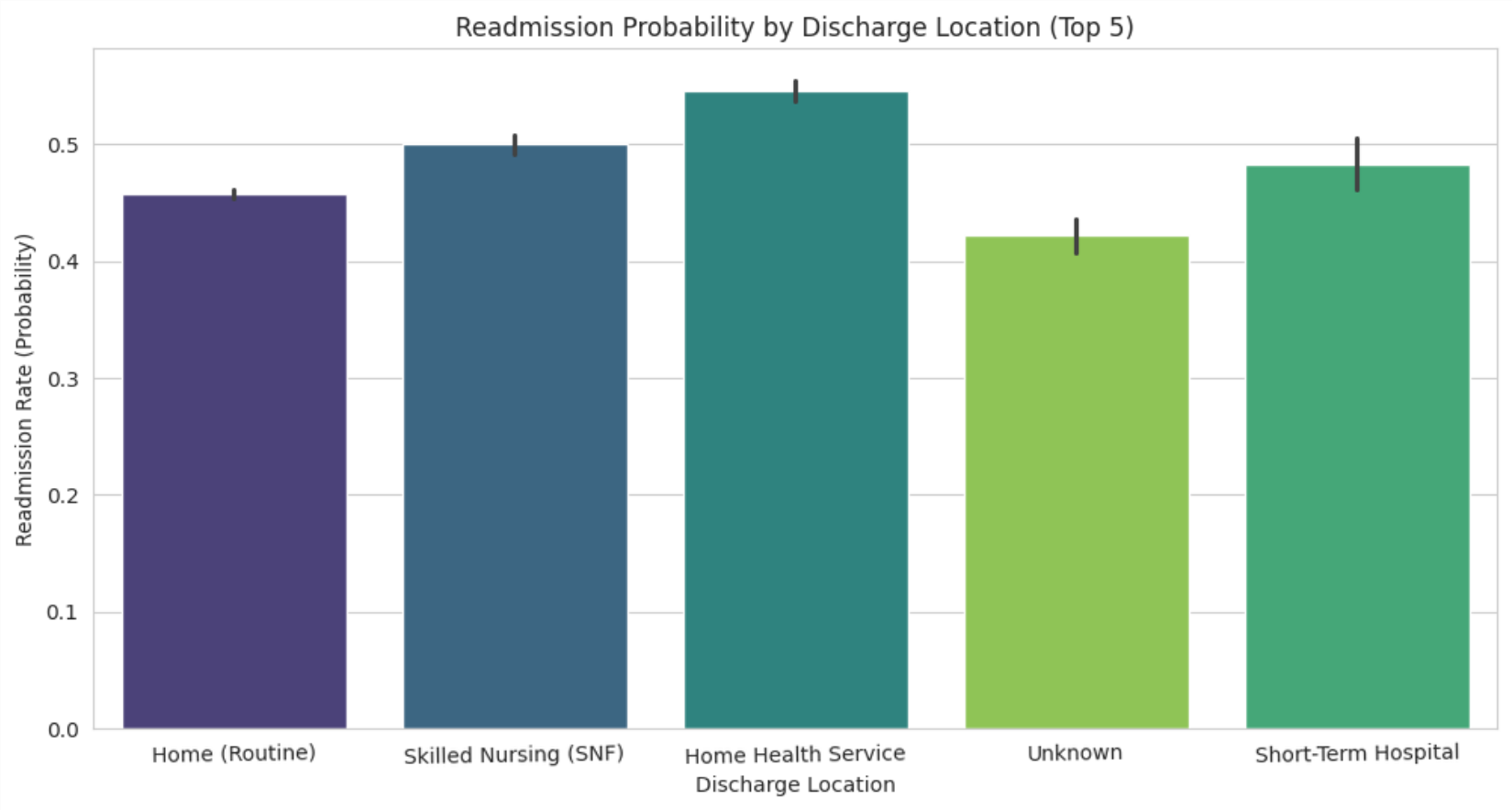
PATIENT SAFETY



No Correlation: Extending the length of stay does **not** statistically reduce readmission risk.

The Failure of the Safety Net

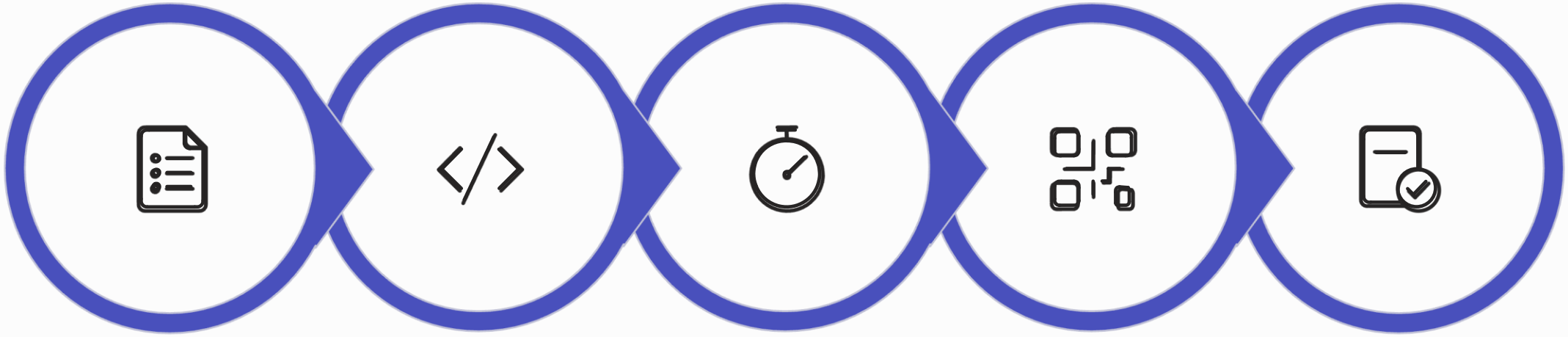
When the safety net becomes a danger zone



Critical Finding: We are discharging our most fragile patients into environments where support where support infrastructure is systematically failing. This represents our highest-priority priority intervention opportunity.

Diagnostic Enrichment

Automated Data Pipeline



Target List **Request** **1-Second Delay Delay** **HTML Parsing Parsing** **Enriched Data**

This robust architecture ensures efficient and ethical data collection, transforming raw web data into actionable insights for patient risk stratification. We targeted the top 20 most frequent codes and fetched their standard medical descriptions.

From Cryptic Codes to Clinical Context

BEFORE

The Problem: Raw ICD-9 Data

428

38

250.06

486

AFTER

The Solution: Enriched Diagnostics

Heart failure

Septicemia

Diabetes with neurological manifestations

486 Pneumonia, organism unspecified

Transforming raw administrative codes into actionable clinical intelligence.

VCI Model & Recommendations

Moving from Analysis to Action

The Vitality Complexity Index (VCI) - Adapted from the industry-standard LACE Index

$$VCI = [L]ength\ of\ Stay + [A]cuity\ of\ Admission + [C]ombordities\ Burden + [E]mergency\ Visit\ Intensity\ Score$$

Length of Stay (L)

Duration of hospital visit, reflecting initial severity and recovery needs.

Acuity (A)

Severity of illness at admission, indicating critical care requirements.

Comorbidities (C)

Number and complexity of existing chronic health conditions.

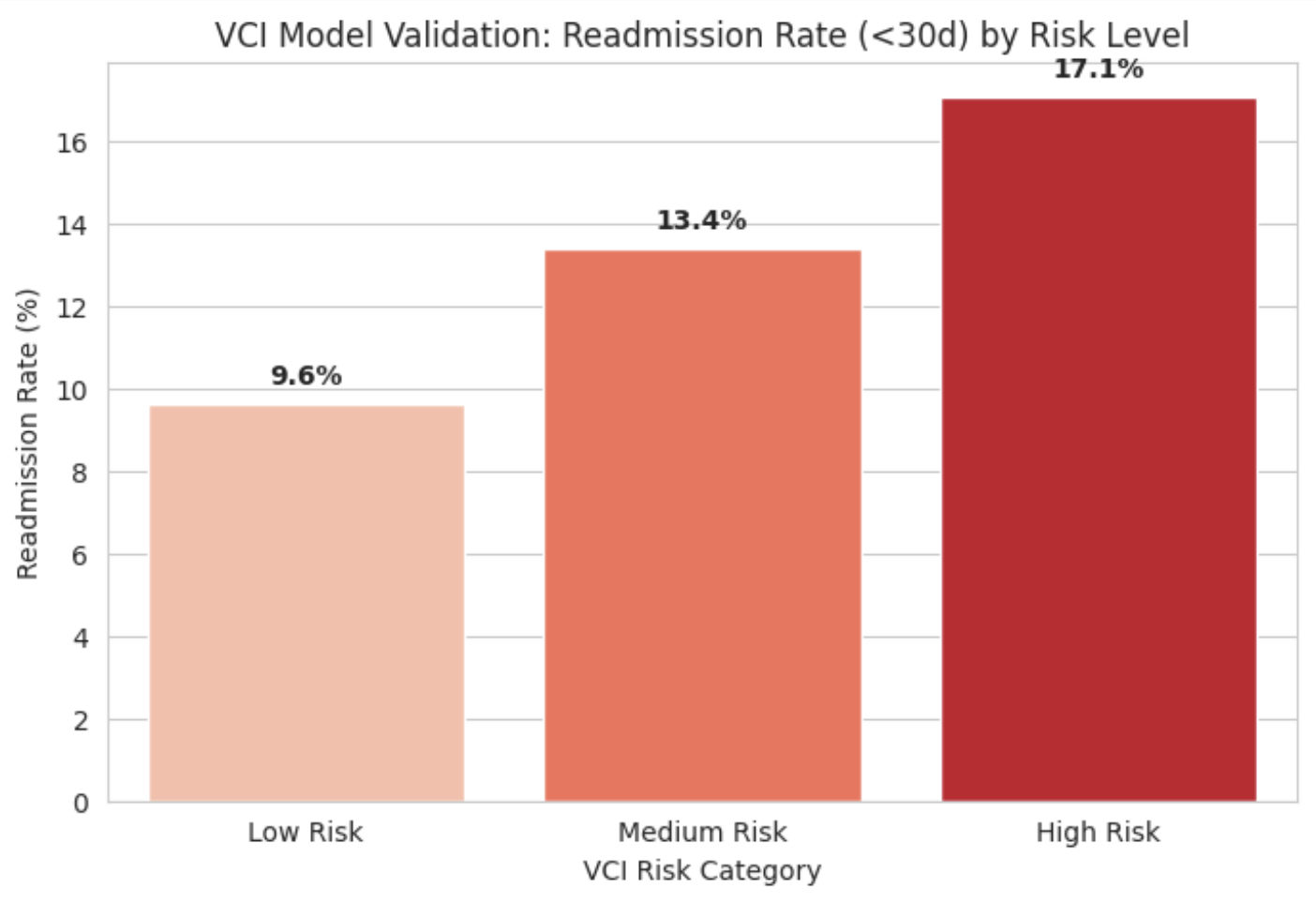
Emergency History (E)

Prior emergency department visits, signaling potential instability.

Adapted from the industry-standard LACE Index, the VCI provides a quantifiable measure of patient risk.

Validated Predictive Power

Model Validation (Risk Stratification)



Our predictive model effectively stratifies patients into distinct risk cohorts for readmission within 30 days:

- **Low Risk (<7):** Demonstrates a a baseline readmission rate of of approximately **9%**.
- **Medium Risk (7-10):** Shows an elevated rate of approximately **13%**.
- **High Risk (>10):** Presents a significantly higher readmission probability of **17.1%**.

The High Risk group returns at **1.8x** the rate of the Low Risk group, validating the model's ability to identify vulnerable patients.

Strategic Recommendations

Three evidence-based interventions to reduce readmissions and mitigate financial risk

Digital Triage Integration

Integrate Vitality Complexity Index (VCI) into clinical dashboard with automated alerts for scores >10, enabling proactive intervention at discharge

Home Health Audit Protocol

Mandate 24-hour follow-up contact contact for all Home Health discharges to address the readmission failure rate through early intervention

Insulin Mastery Program

Implement mandatory "Teach-Back" education track for insulin-dependent patients, replacing inadequate 15-minute protocols with competency-based discharge criteria

Thank You!